

St George Girls High School

Trial Higher School Certificate Examination

2007



Mathematics Extension 1

Total Marks – 84

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Attempt ALL questions.
- Begin each question on a new booklet
- Write your student number on each page
- All necessary working must be shown.
- Diagrams are not to scale.
- Board-approved calculators may be used.
- The mark allocated for each question is listed at the side of the question.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – 12 marks (Start a new booklet)

Marks

- a) Differentiate $\log(xe^x)$ 2
- b) Find the equation of the normal on the curve $y = \ln(x+2)$ at the point $(0, \ln 2)$ 3
- c) Solve $\frac{2x+3}{x-4} \geq 1$ 2
- d) Let A be the point $(3, -1)$ and B be the point $(9, 2)$. Find the coordinates of the point P which divides the interval AB externally in the ratio $5:2$ 2
- e) Evaluate the limit $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{2x}$ 2

Question 2 – 12 marks (Start a new booklet)

Marks

- a) Find $\int \frac{19dx}{4+8x^2}$ 2
- b) Find the acute angle between the lines $x+2y=5$ and $x-3y=-3$ 2
- c) Find $\frac{d}{dx}(\cos^{-1}(2\cos^2 x - 1))$ in simplest terms for $0 \leq x \leq \frac{\pi}{2}$ 2
- d) Evaluate $\int_0^2 \frac{2x}{\sqrt{x^2+1}} dx$ by using the substitution $u = x^2 + 1$ 4
- e) Differentiate $x^2 \tan 5x$ 2

Question 3 – 12 marks (Start a new booklet)

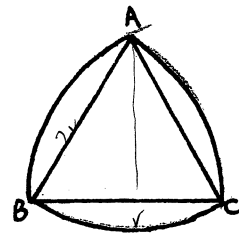
Marks

- a) Write down the period and amplitude of $y = 2 \cos \frac{1}{3}x$ 2
- b) Use the change of base formula to evaluate $\log_3 14$ correct to one decimal place. 1
- c) (i) Write $\sqrt{3} \sin x - \cos x$ in the form $r \sin(x - \alpha)$ 0
- (ii) And hence or otherwise solve $\sqrt{3} \sin x - \cos x = 1$ $0 \leq x \leq 2\pi$ 2
- d) Sketch a graph of $y = \frac{3}{\pi} \cos^{-1} \frac{x}{2}$ indicating its domain and range. 4
- e) Find $\frac{d}{dx} (2 \sin^{-1} x)$ 0

Question 4 – 12 marks (Start a new booklet)

Marks

- a) ABC is an equilateral triangle, side $2r$.
The circular arcs AB , BC and CA have centres at C , A and B respectively.



Show that for the figure bounded by the arcs:

- (i) The perimeter is equal to that of a circle of radius r 2
- (ii) The area is approximately 90% of that of a circle, radius r 3
- b) Use the principle of mathematical induction to prove that $3^{2n} - 1$ is divisible by 8 when n is a positive integer. 4
- c) Given that $f'(x) = 1 - \frac{2}{x}$ and the graph of $y = f(x)$ passes through the point $(e, -2)$ find $f(x)$ 3

Question 5 – 12 marks (Start a new booklet)

Mark

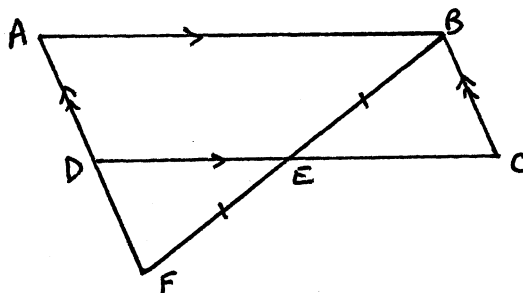
- a) The remainder when $x^3 + ax^2 - 3x + 5$ is divided by $(x + 2)$ is 11.
 Find the value of a .

2

- b) Find the general solution of $2\cos x + \sqrt{3} = 0$

2

- c) In the diagram below $ABCD$ is a parallelogram, $BE = EF$ and AD is produced to F .



- (i) Prove that $\triangle DEF$ is congruent to $\triangle BEC$

- (ii) Hence prove that $DE = \frac{1}{2}DC$

- d) If $y = \frac{x}{\operatorname{cosec} x}$ find $\frac{dy}{dx}$

1

- e) Given that $\log_b \left(\frac{p}{q} \right) = 3$ and $\log_b \left(\frac{p}{r} \right) = 1.6$

2

Find $\log_b \left(\frac{p}{r} \right)$

Question 6 – 12 marks (Start a new booklet)

Marks

- a) Use Newton's method to find a second approximation to the positive root of $x - 2\sin x = 0$. Take $x = 1.7$ as the first approximation. 2
- b) One of the roots of the equation $x^3 + 6x^2 - x - 30 = 0$ is equal to the sum of the other two roots. Find the value of the three roots. 3
- c) A spherical balloon is being inflated and its radius is increasing at a constant rate of 3cm/min. At what rate is its volume increasing when the radius of the balloon is 5cm? 3
- d) From a point P due south of a vertical tower, the angle of elevation of the top of the tower is 20° and from a point Q due east of the tower it is 35° . If the distance from P to Q is 40 metres, find the height of the tower. 4

Question 7 – 12 marks (Start a new booklet)

Marks

- a) A particle is moving in simple harmonic motion. Its displacement x metres at any time t seconds is given by $x = 3 \cos(2t + 5)$

6

- (i) Find the period and amplitude of the motion.
- (ii) Find the maximum acceleration of the particle.
- (iii) Find the speed of the particle when $x = 2$

- b) A projectile is fired with initial velocity V m/s at an angle of projection θ from a point O on horizontal ground. After 2 seconds it just passes over a 10 metre high wall that is 12 metres from the point of projection.

6

Assume acceleration due to gravity is 10m/sec^2 . Assume the equations of displacement are $x = Vt \cos \theta$ and $y = -5t^2 + Vt \sin \theta$

- (i) Find V and θ to the nearest degree.
- (ii) Find the maximum height reached by the projectile.
- (iii) Find the range in the horizontal plane through the point of projection.

End of Paper

TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

Note $\ln x = \log_e x, \quad x > 0$

Trial Solutions Mathematics 2007

Question 1

$$\begin{aligned} (a) \quad \frac{d}{dx} \log(xe^x) &= \frac{d}{dx} (\ln x + \ln e^x) \\ &= \frac{d}{dx} (\ln x + x) \\ &= \frac{1}{x} + 1 \end{aligned}$$

$$(b) \quad y = \ln(x+2)$$

$$y' = \frac{1}{x+2}$$

$$x=0 \quad y' = \frac{1}{2}$$

$$\therefore m_T = \frac{1}{2} \quad m_N = -2$$

$$\therefore m = -2 \quad (0, \ln 2)$$

$$y - y_1 = m(x - x_1)$$

$$y - \ln 2 = -2(x - 0)$$

$$y - \ln 2 = -2x$$

$$2x + y - \ln 2 = 0$$

$$(c) \quad \frac{2x+3}{x-4} \geq 1 \quad x \neq 4$$

$$(x-4)(2x+3) \geq (x-4)^2$$

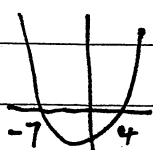
$$2x^2 - 5x - 12 \geq x^2 - 8x + 16$$

$$x^2 + 3x - 28 \geq 0$$

$$(x+7)(x-4) \geq 0$$

$$x > 4$$

$$x \leq -7$$



$$\begin{array}{ccc} & A & B \\ (3, -1) & & (9, 2) \end{array} \quad -5:2$$

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{-5 \cdot 9 + 2 \cdot 3}{-5 + 2}, \frac{-5 \cdot 2 + 2 \cdot -1}{-5 + 2} \right)$$

$$= (13, 4)$$

$$(e) \quad \lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{2} \cdot \frac{1}{5} \cdot \frac{\sin \frac{x}{5}}{\frac{x}{5}}$$

$$= \frac{1}{10} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{\frac{x}{5}}$$

$$= \frac{1}{10}$$

Question 2

$$\begin{aligned}
 (a) \int \frac{19 dx}{4+8x^2} &= \frac{19}{8} \int \frac{dx}{\frac{1}{2}+x^2} \\
 &= \frac{19}{8} \cdot \frac{1}{\frac{1}{\sqrt{2}}} \cdot \tan^{-1} \frac{x}{\frac{1}{\sqrt{2}}} + C \\
 &= \frac{19\sqrt{2}}{8} \tan^{-1} \sqrt{2}x + C
 \end{aligned}$$

$$\begin{aligned}
 (b) \tan \alpha &= \frac{m_1 - m_2}{1 + m_1 m_2} \\
 x+2y &= 5 & x-3y &= -3 \\
 2y &= -x+5 & 3y &= x+3 \\
 y &= -\frac{1}{2}x + \frac{5}{2} & y &= \frac{1}{3}x + 1
 \end{aligned}$$

$$\therefore m_1 = -\frac{1}{2} \quad m_2 = \frac{1}{3}$$

$$\tan \alpha = \frac{-\frac{1}{3} - \frac{1}{3}}{1 + (-\frac{1}{2}) \cdot \frac{1}{3}}$$

$$= \frac{-\frac{2}{3}}{\frac{5}{6}}$$

$$= -1$$

$$\therefore \tan \alpha = -1$$

$$\alpha = 135^\circ$$

$$\therefore \text{acute angle} = 45^\circ$$

$$(d) \frac{d}{dx} (\cos^{-1} (2 \cos^2 x - 1))$$

$$= \frac{d}{dx} (\cos^{-1} (\cos 2x))$$

$$= \frac{d}{dx} (2x)$$

$$= 2$$

$$(d) \int_0^2 \frac{2x}{\sqrt{x^2+1}} dx$$

$$\text{let } u = x^2 + 1 \quad x=2 \quad u=5$$

$$du = 2x dx \quad x=0 \quad u=1$$

$$\therefore I = \int_1^5 \frac{du}{\sqrt{u}}$$

$$= \int_1^5 u^{-\frac{1}{2}} du$$

$$= \left[\frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^5$$

$$= 2u^{\frac{1}{2}} \Big|_1^5$$

$$= 2\sqrt{5} - 2\sqrt{1}$$

$$= 2\sqrt{5} - 2$$

$$(e) y = x^2 \tan 5x$$

$$y' = 2x \tan 5x + x^2 \cdot 5 \sec^2 5x$$

$$= 2x \tan 5x + 5x^2 \sec^2 5x$$

Question 3

(a) $y = 2 \cos \frac{1}{3} x$

period = $\frac{2\pi}{\frac{1}{3}}$

amp = 2

(b) $\log_3 14 = \frac{\log 14}{\log 3}$
 $= 2.4$

(c)(i) $\sqrt{3} \sin x - \cos x$

$$r \sin(x - \alpha) = r (\sin x \cos \alpha - \cos x \sin \alpha)$$

$$= r \cos \alpha \sin x - r \sin \alpha \cos x$$

$\therefore \sqrt{3} \sin x - \cos x = r \cos \alpha \sin x - r \sin \alpha \cos x$

$r \cos \alpha = \sqrt{3}$ $r^2 \cos^2 \alpha = 3$

$r \sin \alpha = 1$ $r^2 \sin^2 \alpha = 1$

$r^2 = 4$

$r = 2$

$\tan \alpha = \frac{1}{\sqrt{3}}$

$\alpha = \frac{\pi}{6}$

$\therefore \sqrt{3} \sin x - \cos x = 2 \sin \left(x - \frac{\pi}{6}\right)$

(ii) $\sqrt{3} \sin x - \cos x = 1$

$\therefore 2 \sin \left(x - \frac{\pi}{6}\right) = 1$

$\sin \left(x - \frac{\pi}{6}\right) = \frac{1}{2}$

$x - \frac{\pi}{6} = \frac{\pi}{3}$ $\frac{2\pi}{3}$

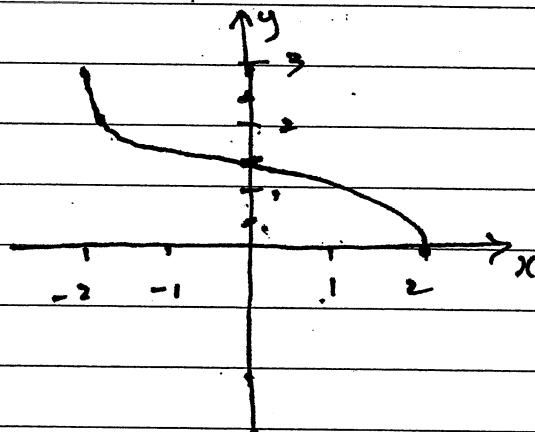
(d) $y = \frac{3}{\pi} \cos^{-1} \frac{x}{2}$

$-1 \leq \frac{x}{2} \leq 1$

$-2 \leq x \leq 2$

$0 \leq \cos^{-1} \frac{x}{2} \leq \pi$

$0 \leq \frac{3}{\pi} \cos^{-1} \frac{x}{2} \leq 3$



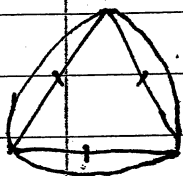
domain $-2 \leq x \leq 2$

range $0 \leq y \leq 3$

(e) $\frac{d}{dx} (2 \sin^{-1} x) = 2 \cdot \frac{1}{\sqrt{1-x^2}}$
 $= \frac{2}{\sqrt{1-x^2}}$

Question 4

(a) i



$$60^\circ = \frac{\pi}{3}$$

length of one arc

$$\text{radius} = 2r$$

$$\theta = \frac{\pi}{3}$$

arc length = $r\theta$

$$= 2\pi \cdot \frac{\pi}{3}$$

$$3 \text{ arc lengths} = 3 \cdot \frac{2\pi r}{3}$$

$$= 2\pi r$$

(ii) Area of one segment

$$= \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} (2r)^2 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$$

$$= \frac{1}{2} \cdot 4r^2 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$= r^2 \left(\frac{2\pi}{3} - \sqrt{3} \right)$$

$$\text{Area of } \Delta = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \cdot 2r \cdot 2r \cdot \sin \frac{\pi}{3}$$

$$= 2r^2 \cdot \frac{\sqrt{3}}{2}$$

$$= r^2 \sqrt{3}$$

$$\text{Total area} = \sqrt{3} r^2 + 3 \left[r^2 \left(\frac{2\pi}{3} - \sqrt{3} \right) \right]$$

$$= \sqrt{3} r^2 + 3r^2 \left(\frac{2\pi}{3} - \sqrt{3} \right)$$

$$= r^2 \left[3 \left(\frac{2\pi}{3} - \sqrt{3} \right) + \sqrt{3} \right]$$

$$= r^2 (2\pi - 2\sqrt{3})$$

$$\% \text{ of circle} = \frac{r^2 (2\pi - 2\sqrt{3})}{\pi r^2} \times 100\%$$

$$= 89.7\%$$

$$\approx 90\%$$

(b)

Show true for $n=1$

$$3^2 - 1 = 8 \quad \therefore \text{divisible by 8}$$

let statement be true for $n=k$

$$\therefore 3^{2k} - 1 = 8M$$

Need to show true for $n=k+1$

$$3^{2(k+1)} - 1 = 3^{2k+2} - 1$$

$$= 3^{2k} \cdot 3^2 - 1$$

$$= 3^{2k} \cdot 9 - 1$$

$$= 3^{2k} \cdot 9 - 9 + 8$$

$$= 9(3^{2k} - 1) + 8$$

$$= 9 \cdot 8M + 8$$

$$= 8(9M + 1)$$

$\therefore$ divisible by 8

Since true for $n=1$ then

must be true for $n=k+1=2$

then $n=2+1=3 \therefore$ true for all n

$n \geq 1$

$$(c) f(x) = 1 - \frac{2}{x}$$

$$f(x) = \int 1 - \frac{2}{x} dx$$

$$= x - 2 \ln x + C$$

$$y = -2 \text{ when } x = e$$

$$-2 = e - 2 \ln e + C$$

$$C = e$$

$$\therefore f(x) = x - 2 \ln x + e$$

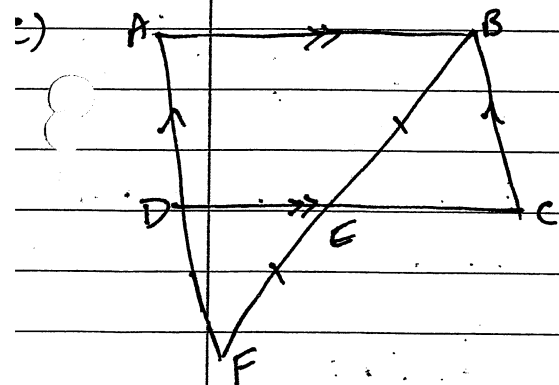
Question 5

(a) $P(x) = x^3 + ax^2 - 3x + 5$
 $P(-2) = 11$
 $P(-2) = (-2)^3 + a(-2)^2 - 3(-2) + 5$
 $= -8 + 4a + 6 + 5$
 $= 3 + 4a$
 $\therefore 3 + 4a = 11$
 $4a = 8$
 $a = 2$

(b) $2 \cos x = -\sqrt{3}$
 $\cos x = -\frac{\sqrt{3}}{2}$
 $x = \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

general solution

$$x = 2n\pi \pm \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$$



$\angle DEF = \angle BEC$ (vertically op. angles)
 $\angle BCE = \angle EDF$ (alternate angles on parallel lines)
 $FE = BE$ (given)

$\therefore \triangle DEF \equiv \triangle CEB$ (AAS)

(ii) $DE = EC$ (corresponding sides in congruent Δ s)

$\therefore E$ is midpoint of DC

$\therefore DE = \frac{1}{2} DC$

(d) $y = \frac{x}{\operatorname{cosec} x}$

$y = x \sin x$

$y' = \sin x + x \cos x$

(e) $\log\left(\frac{P}{Q}\right) = 3$ $\log\left(\frac{Q}{R}\right) = 1.6$

$\frac{P}{R} = \frac{P}{Q} \cdot \frac{Q}{R}$

$\therefore \log\left(\frac{P}{R}\right) = \log\left(\frac{P}{Q} \cdot \frac{Q}{R}\right)$

$= \log \frac{P}{Q} + \log \frac{Q}{R}$

$= 3 + 1.6$

$= 4.6$

Question b.

(a) $x - 2\sin x = 0$ $f(x) = x - 2\sin x$
 $f'(x) = 1 - 2\cos x$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f(1.7) = -0.2833$$

$$f'(1.7) = 1.2577$$

$$= 1.7 - \frac{-0.2833}{1.2577}$$

$$= 1.9253$$

(b) $x^3 + 6x^2 - x - 30 = 0$

$$\alpha, \beta, \alpha + \beta$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha + \beta + \alpha + \beta = -6$$

$$2\alpha + 2\beta = -6$$

$$\alpha + \beta = -3$$

$$\alpha = -3 - \beta$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta + \alpha(\alpha + \beta) + \beta(\alpha + \beta) = -1$$

$$\alpha\beta + (\alpha + \beta)^2 = -1$$

$$\alpha\beta + 9 = -1$$

$$\alpha\beta = -10$$

$$\therefore \beta = -5, 2$$

$\therefore$ the three roots

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$\alpha\beta(\alpha + \beta) = 30$$

$$\therefore \alpha, \beta, \gamma$$

$$\alpha\beta = -10$$

$$-5, 2, -3$$

$$(-3 - \beta)\beta = -10$$

$$\beta^2 + 3\beta - 10 = 0$$

$$(\beta + 5)(\beta - 2) = 0$$

(c) $\frac{dV}{dt} = ?$ $r = 5$ $\frac{dr}{dt} = 3$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

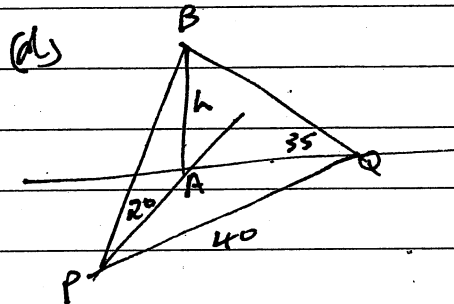
$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$= 4\pi r^2 \cdot \frac{dr}{dt}$$

$$r = 5$$

$$\frac{dV}{dt} = 4 \cdot \pi \cdot 5^2 \cdot 3$$

$$= 300\pi \text{ cm}^3/\text{min}$$



$$\tan 20 = \frac{h}{PA}$$

$$\tan 35 = \frac{h}{QA}$$

$$PA = \frac{h}{\tan 20}$$

$$QA = \frac{h}{\tan 35}$$

$$PA^2 + QA^2 = 40^2$$

$$\left(\frac{h}{\tan 20}\right)^2 + \left(\frac{h}{\tan 35}\right)^2 = 1600$$

$$h^2 \left(\frac{1}{\tan^2 20} + \frac{1}{\tan^2 35}\right) = 1600$$

$$h = \sqrt{\frac{1600}{\frac{1}{\tan^2 20} + \frac{1}{\tan^2 35}}}$$

$$= \sqrt{166.87}$$

$$= 12.9 \text{ m}$$

Question 7

(a) $x = 3 \cos(2t + 5)$

(i) period = $\frac{2\pi}{\omega}$ amplitude = 3
 $= \frac{2\pi}{2}$
 $= \pi$

(ii) $x = 3 \cos(2t + 5)$
 $\dot{x} = -6 \sin(2t + 5)$
 $\ddot{x} = -12 \cos(2t + 5)$
max $\cos(2t + 5) = -1$
 $\ddot{x} = 12 \text{ m/s}^2$

(iii) $x = 2$
 $3 \cos(2t + 5) = 2$
 $\cos(2t + 5) = \frac{2}{3}$
 $2t + 5 = 5.44$
 $2t = 0.44$
 $t = 0.22$

$\dot{x} = -6 \sin(2 \times 0.22 + 5)$
 $= -6 \sin 5.44$
 $= 4.47 \text{ m/s}$

b) $x = V \cos \theta \cdot t$ $y = V \sin \theta \cdot t - 5t^2$

i) $t = 2$ $y = 10$ $t = 2$ $x = 12$
 $12 = 2V \cos \theta$
 $10 = 2V \sin \theta - 20$ $\therefore \tan \theta = \frac{30}{12}$
 $30 = 2V \sin \theta$
 $\frac{2V \sin \theta}{2V \cos \theta} = \frac{30}{12}$
 $\theta = 68.12^\circ$

$V \cos \theta = 6$ $V^2 \cos^2 \theta = 36$

$V \sin \theta = 15$ $V^2 \sin^2 \theta = 225$

$V^2 (\sin^2 \theta + \cos^2 \theta) = 261$

$V^2 = 261$

$V = \sqrt{261}$

$= 16.6 \text{ m/s}$

(ii)

$\ddot{y} = -10$

$\dot{y} = \int -10 dt$

$\dot{y} = -10t + c$

$y = V \sin \theta \cdot t - 5t^2$

$\therefore \dot{y} = -10t + V \sin 68.12^\circ$

$y = \int -10t + V \sin \theta$

$y = -5t^2 + V \sin \theta \cdot t$

$y = 0$ $t = 0$ $c = 0$

$\therefore y = -5t^2 + V \sin \theta$

$\dot{y} = 0$ for max height

$-10t + V \sin \theta = 0$

$t = \frac{V \sin \theta}{10}$

$y = V \sin \theta \cdot t - 5t^2$

$= V \sin \theta \cdot \frac{V \sin \theta}{10} - 5 \left(\frac{V \sin \theta}{10} \right)^2$

$= \frac{V^2 \sin^2 \theta}{10} - 5 \frac{V^2 \sin^2 \theta}{100}$

$= \frac{\sqrt{2} \sin^2 \theta}{20}$

but $V = 16.16$ $\theta = 68.12^\circ$

$\therefore y = \frac{(16.16)^2 \sin^2 68.12^\circ}{20}$

$= 11.25 \text{ m}$

(iii) time of flight = $\frac{2t}{1}$
 $= \frac{V \sin \theta}{5}$

range

$x = V \cos \theta \cdot t$

$= V \cos \theta \cdot \frac{V \sin \theta}{5}$

$= \frac{V^2 \cos \theta \cdot \sin \theta}{5}$

$= 17.99 \approx 18 \text{ m}$